

CLINICAL TRIAL PROTOCOL

Protocol ID: SHAMNA-2026-027

ClinicalTrials.gov Identifier: NCT07618234

Study Title: Dermocosmetic Evaluation of Propolis Ointments in Atopic-Prone Dry Skin: An Exploratory Quadruple-Blind, Randomized, Vehicle-Controlled, Parallel-Group Pilot Study

1. ADMINISTRATIVE INFORMATION

1.1 Protocol Identification

- Protocol Number: SHAMNA-2026-027
- Version: 3.0
- Date: April 1, 2026
- Amendment: Version 4.0 (dated [25-05-2026], before database lock)

1.2 Regulatory Identifiers

- ClinicalTrials.gov: NCT07618234
- Ethics Committee: SHAMNA Biomedical Ethics Committee
- Ethics Approval: SHAMNA-2026-027
- Institution: Manara University, Latakia, Syria

1.3 Principal Investigator

- Name: Chadi Khatib, PhD
- Department: Pharmacognosy, Faculty of Pharmacy
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1.4 Study Team

- Co-Investigators: Mahmoud Bitar, Haya Farhat, Nagham Saleh
- Institution: Faculty of Pharmacy, Manara University, Syria

2. SYNOPSIS

2.1 Study Design

- Type: Interventional, exploratory pilot study
- Allocation: Randomized (1:1:1)

- Intervention Model: Parallel assignment
- Masking: Quadruple-blind (participants, care providers, investigators, assessors)
- Control: Vehicle-controlled
- Purpose: Treatment (feasibility and exploratory signal detection)

2.2 Study Population

- Target Condition: Atopic-prone xerotic skin (dry skin with atopic tendency)
- Sample Size: n = 30 (10 per group)
- Recruitment: Single-center (Syrian Scientific Society for Herbs & Alternative Medicines & Nutrition, Aleppo, Syria)
- Duration: April 15 – May 24, 2026 (4 weeks treatment + screening)

2.3 Interventions

- Group A: EEP Ointment 3% (Ethanollic Extract of Propolis)
- Group B: Crude Propolis Ointment 5%
- Group C: Vehicle Ointment (base control)

2.4 Primary Objective

Feasibility assessment and exploratory signal detection for dermocosmetic supportive benefits of propolis-based formulations in atopic-prone xerotic skin.

2.5 Primary Outcome

Change in Clinical Dryness Score (5-point ordinal scale: 0=none to 4=very severe) from Baseline (Day 0) to Week 4.

2.6 Secondary Outcomes

- Pruritus VAS (0-10)
- Skin Comfort (Likert 1-5)
- Cosmetic Acceptability
- Subject Satisfaction (Week 4 only)
- Safety and tolerability

2.7 Exploratory Outcomes

- Localized SCORAD (included in amended SAP v2.0, before database lock)
- Marked with † in all figures and tables

3. BACKGROUND AND RATIONALE

3.1 Condition Background

Atopic-prone xerotic skin represents a prevalent dermatologic concern characterized by impaired epidermal barrier function, transepidermal water loss (TEWL), and susceptibility to inflammatory flares. In Syria, fixed-dose topical combination creams containing corticosteroids are frequently dispensed without prescription, contributing to unrecognized corticosteroid exposure and potential topical steroid damaged facies (TSDF).

3.2 Propolis Rationale

Propolis (bee glue) possesses antimicrobial, anti-inflammatory, and antioxidant properties. Previous clinical investigations have primarily evaluated ethanolic extracts (EEP) rather than crude resin traditionally applied with native beeswax. The "propolis-beeswax synergy" hypothesis has not been previously investigated clinically.

3.3 Study Hypotheses

H1: Crude propolis will demonstrate comparable exploratory supportive benefits to EEP.

H2: Both formulations will significantly outperform vehicle in exploratory clinical outcomes, supporting their feasibility as non-steroidal supportive adjuncts within steroid-sparing dermatologic care.

4. STUDY DESIGN AND METHODOLOGY

4.1 Design Overview

This exploratory, quadruple-blind, randomized, vehicle-controlled, parallel-group pilot study was conducted at Syrian Scientific Society for Herbs & Alternative Medicines & Nutrition, Aleppo, Syria in cooperation with Manara University, April 15 – May 24, 2026.

4.2 Ethical Considerations

- Approved by SHAMNA Biomedical Ethics Committee (SHAMNA-2026-027)
- Registered on ClinicalTrials.gov (NCT07618234)
- Conducted in accordance with Declaration of Helsinki (2024 revision)
- Good Clinical Practice (GCP) principles
- CONSORT 2010 pilot guidelines (Eldridge et al., 2016)

4.3 Informed Consent

All participants provided written informed consent in Arabic.

Consent forms available in Arabic and English (Documents S4).

5. STUDY POPULATION

5.1 Inclusion Criteria

1. Adults aged 18-60 years
2. Atopic-prone xerotic skin (dryness score ≥ 2 on 5-point scale)
3. Willingness to abstain from other topical products

4. Ability to comply with study procedures

5.2 Exclusion Criteria

1. Active skin infection
2. Allergy to propolis/bee products/ointment components
3. Systemic corticosteroids/immunosuppressants within 4 weeks
4. Pregnancy or lactation
5. Participation in other clinical trial within 30 days

5.3 Withdrawal Criteria

1. Severe adverse reaction
2. Worsening condition requiring rescue therapy
3. Poor compliance (<80% adherence)
4. Participant request
5. Investigator discretion for safety

6. INTERVENTIONS

6.1 Formulation Preparation

All formulations prepared aseptically under Class II laminar flow hood (Esco Airstream, Singapore) using Ph.Eur. grade components (Fagron, Rotterdam).

6.2 Formulation A — EEP Ointment 3% (w/w)

- EEP 3.0%
- White soft paraffin 67.0%
- Liquid paraffin 20.0%
- Anhydrous lanolin 10.0%

6.3 Formulation B — Crude Propolis Ointment 5% (w/w)

- Micronized crude propolis 5.0%
- White soft paraffin 65.0%
- Liquid paraffin 20.0%
- Anhydrous lanolin 10.0%

6.4 Formulation C — Vehicle Ointment

- White soft paraffin 70.0%
- Liquid paraffin 20.0%
- Anhydrous lanolin 10.0%

6.5 Manufacturing Process

- Melting at 70°C ($\pm 1^\circ\text{C}$, Julabo TW20, Germany)
- Mixing under mechanical stirring (500 rpm, 30 min, IKA RW 20 digital)
- Cooling to 20-25°C
- Filling into identical 30g aluminum tubes
- Coded by independent pharmacist (A, B, C)
- Storage at 20-25°C protected from light

6.6 Blinding Verification

- Similarity $> 8.5/10$ (n=10 naive assessors)
- Indistinguishable by color, odor, texture, viscosity

7. CHEMICAL STANDARDIZATION

7.1 Propolis Source

- Location: Jableh district, Latakia Governorate, Syria
- Coordinates: 35.3223593°N, 35.9397543°E
- Altitude: 43m above sea level
- Collection: Mediterranean coastal region

7.2 HPLC Fingerprinting

- Similarity index ≥ 0.92 vs. 2020 regional reference profile (Khatib et al., 2020)
- Identified markers:
 - CAPE (18.4 min)
 - Pinocembrin (19.8 min)
 - Galangin (22.1 min)
 - Chrysin (24.7 min)
 - Artepillin C (26.3 min)

7.3 Total Phenolic Content

- 287.4 ± 11.2 mg GAE/g

7.4 Antioxidant Activity (DPPH)

- IC₅₀: 82.3 $\mu\text{g/mL}$
- Reference value: 79.71 $\mu\text{g/mL}$ (Khatib et al., 2020)

7.5 Particle Size Analysis

- D50 = $28.4 \pm 8.2 \mu\text{m}$ (Mastersizer 3000, Malvern Panalytical)
- Reproducibility CV <5% across three independent batches

8. STUDY PROCEDURES

8.1 Randomization

- 1:1:1 allocation
- Block size 6
- Stratified by sex
- Computer-generated by independent statistician
- Allocation concealment: sequentially numbered, opaque, sealed envelopes (SNOSE)

8.2 Blinding

- Quadruple-blind: participants, care providers, investigators, assessors
- Unblinding permitted only in medical emergencies

8.3 Intervention Administration

- Dose: 2 mg/cm^2
- Application site: flexor forearm ($10 \times 10 \text{ cm}$)
- Frequency: twice daily for 4 weeks
- Compliance: assessed by tube weight and dosing diary

8.4 Visit Schedule

- Visit 0 (Day 0): Baseline — screening, randomization, baseline assessment
- Visit 1 (Week 2, Day 14 ± 2): Safety and efficacy evaluation
- Visit 2 (Week 4, Day 28 ± 2): Final assessment

8.5 Permitted Concomitant Medications

- Mild soap
- Fragrance-free moisturizer (if needed)

8.6 Prohibited Medications

- Topical corticosteroids
- Calcineurin inhibitors
- Herbal products
- Medicated creams (except rescue therapy)

8.7 Rescue Therapy

- Hydrocortisone 1% permitted only for severe flare
- Must be documented (dose, duration, area)

9. OUTCOME MEASURES

9.1 Primary Outcome

Clinical Dryness Score (5-point ordinal scale):

0 = None

1 = Mild

2 = Moderate

3 = Severe

4 = Very severe

Assessment: Change from Baseline to Week 4

Assessor: Independent blinded dermatologist

Conditions: LED 5500K, 22±2°C, 10:00-12:00 AM

9.2 Secondary Outcomes

- Pruritus VAS: 0-10 scale
- Skin Comfort: Likert 1-5 scale
- Cosmetic Acceptability: Assessed at Week 4
- Subject Satisfaction: Week 4 only, Likert scale

9.3 Exploratory Outcomes

- Localized SCORAD: $A/5 + 7B/2 + C/2$
 - Extent: 0-6
 - Intensity: 0-3 per sign (0-12 total)
 - Symptoms: 0-10 each (0-20 total)
- Included in amended SAP v2.0 (before database lock)
- Marked with † in all figures and tables

9.4 Safety Assessments

- Adverse events: recorded at each visit
- Severity: mild, moderate, severe
- Causality: related, possibly related, unrelated
- Serious adverse events: reported to ethics within 24 hours

10. STATISTICAL ANALYSIS

10.1 Analysis Plan

- SAP v1.0: Original plan
- SAP v2.0: Amended (before database lock, includes exploratory SCORAD)

10.2 Software

- Python (v3.x)
- SPSS (v28)
- GraphPad Prism (v10)
- G*Power (v3.1.9.7) for post-hoc power estimation

10.3 Methods

- Normality: Shapiro-Wilk test
- Between-group: Kruskal-Wallis → Mann-Whitney U (Bonferroni-adjusted $\alpha=0.017$)
- Within-group: Friedman → Wilcoxon signed-rank
- Effect size: Cliff's Delta
 - Negligible: $|d| < 0.147$
 - Small: $0.147 \leq |d| < 0.33$
 - Medium: $0.33 \leq |d| < 0.474$
 - Large: $|d| \geq 0.474$
- Missing data: Complete-case analysis (minimal missing)

10.4 Significance

- Two-tailed tests
- $p < 0.05$ (unadjusted)
- $p < 0.017$ (Bonferroni-adjusted post-hoc)

10.5 Sample Size Justification

- Pilot study: $n=30$ (10 per group)
- 100% completion, 0% lost to follow-up
- Post-hoc power estimation for definitive trial design
- Estimated definitive trial: $n \approx 120-150$ per group

11. DATA MANAGEMENT

11.1 Data Collection

- Electronic data capture with unique Study IDs
- De-identified dataset (CSV format)
- Source data verification

11.2 Data Storage

- Secure electronic storage at Manara University
- Backup procedures
- Access limited to authorized personnel

11.3 Data Sharing

- Zenodo repository [DOI pending]
- GitHub repository with analysis code
- De-identified raw data for reproducibility
- Additional data available from corresponding author on reasonable request

12. QUALITY ASSURANCE

12.1 Monitoring

- Regular monitoring visits
- Source data verification
- Protocol compliance assessment

12.2 Auditing

- Internal audit by Manara University
- External audit by SHAMNA Ethics Committee

12.3 Documentation

- Case report forms (CRFs)
- Adverse event logs
- Deviation logs
- Amendment records

13. ETHICAL CONSIDERATIONS

13.1 Ethics Committee

- SHAMNA Biomedical Ethics Committee
- Approval: SHAMNA-2026-027

13.2 Regulatory Compliance

- Declaration of Helsinki (2024 revision)
- Good Clinical Practice (ICH-GCP E6 R2)
- CONSORT 2010 Pilot Extension
- SPIRIT 2013 Checklist

13.3 Informed Consent

- Written informed consent in Arabic
- English translation available
- Voluntary participation
- Right to withdraw at any time

13.4 Confidentiality

- Unique Study IDs
- De-identified data
- Secure storage
- Limited access

13.5 Data Protection

- Compliance with local data protection regulations
- No identifying information in publications
- Aggregate reporting only

14. PUBLICATION AND DISSEMINATION

14.1 Authorship

- ICMJE criteria applied
- Contributions documented

14.2 Publication Plan

- Primary manuscript: Q1 dermatology journal
- Target journals: CCID, Dermatologic Therapy, JCD
- Expected: Major Revision → Acceptance

14.3 Data Availability

- Zenodo repository [DOI pending]
- GitHub: Analysis code and protocol
- Supplementary materials: Tables S1-S8, Figures S1-S3

14.4 AI Disclosure

- AI-assisted writing tools used for language refinement
- All scientific content verified by research team
- No AI used for data analysis or statistical computation

15. REFERENCES

- [1] Eldridge SM, Chan CL, Campbell MJ, et al. CONSORT 2010 statement: extension to randomised pilot and feasibility trials. *BMJ*. 2016;355:i5239.
- [2] Khatib C, Aga MIH, Nattouf AH. Chemical Study of the Propolis Collected from The Masyaf Region and Antioxidant Activity Evaluation. *Pharmjournal.asscp*. 2020;6(5):1-12.
- [3] Chan AW, Tetzlaff JM, Gøtzsche PC, et al. SPIRIT 2013 explanation and elaboration: guidance for protocols of clinical trials. *BMJ*. 2013;346:e7586.
- [4] World Medical Association. Declaration of Helsinki: ethical principles for medical research involving human subjects. *JAMA*. 2013;310(20):2191-2194.
- [5] ICH Harmonised Guideline. Integrated Addendum to ICH E6(R1): Guideline for Good Clinical Practice E6(R2). 2016.

16. APPENDICES

Appendix A: Informed Consent Form (Arabic)

Appendix B: Informed Consent Form (English)

Appendix C: Case Report Form (CRF)

Appendix D: Adverse Event Report Form

Appendix E: Randomization Schedule

Appendix F: Blinding Verification Form

Appendix G: Compliance Assessment Form

Appendix H: Rescue Therapy Documentation

17. SIGNATURES

Principal Investigator:

Name: Chadi Khatib, PhD

End Date: 26-05-2026



Name: Mr. Hisham Kasem

Approval Date: 12-04-2026

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